HUJI · 769001 · DYNAMICS AND CODING IN THE BRAIN

Week 02

The Autapse: Fixed Points and Stability

autapse · linear activation · fixed point · perturbation · stability · vector field

CONTENTS

1 Where we are 1
2 The autapse 1
3 The linear autapse 2
4 Fixed points 3
5 Stability 4
6 The full solution 6
7 The geometric view: the vector field 7
8 Connection to Week 1: it is the eigenvalue 7
9 A limitation of the linear model 8
10 Practice problems with full solutions 8
11 Cheat sheet 9

SECTION 01

Where we are

Week 1 ended with the rate model of a network of N neurons [LMLF eq. 2.8-2.9]:

$$\tau \dot{r}_i = -r_i + g(h_i), \qquad h_i = \sum_{j=1}^{N} J_{ij} r_j + h_i^{\text{ext}}.$$

- r_i is the firing rate of neuron i; τ is its time constant ($\sim$10 ms).
- h_i is its total input: what it hears from the other neurons ($J_{ij} r_j$, synapse *from j to i*) plus external input h_i^{ext}.
- g is the activation function: it turns input into a rate.

THE PLAN FOR THIS WEEK

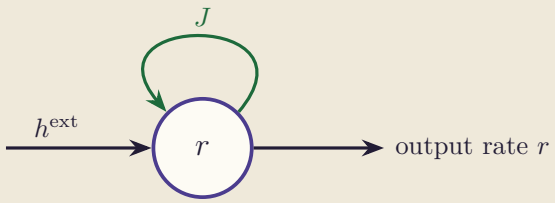
Before analysing big networks, study the smallest possible network: one neuron. If it talks to itself, it already has *feedback*, and feedback is what makes networks interesting. This week we use it to introduce two tools we will use all course long: **fixed points** and their **stability**.

SECTION 02

The autapse

DEF · AUTAPSE

An autapse is a synapse that a neuron makes onto itself: its axon connects back to its own dendrites. A neuron with an autapse is *the simplest possible network*: one neuron, one connection, of strength J [LMLF 2.3.1].



The autapse (after [LMLF Fig. 2.6]). The neuron receives an external input h^{ext}, and its own output r comes back to it through the synapse of strength J.

With a single neuron, the network equations lose their indices [LMLF eq. 2.11-2.12]:

AUTAPSE DYNAMICS

$$\tau \dot{r} = -r + g(h), \qquad h = Jr + h^{\text{ext}}.$$

DEF · EXCITATORY AUTAPSE

We focus on excitatory autapses, $J > 0$: the more the neuron fires, the more input it gives itself. This is positive feedback.

EXAMPLE · ONE INPUT VALUE, TWO SOURCES

$J = 0.5$, $h^{\text{ext}} = 1$, current rate $r = 4$: total input $h = 0.5 \cdot 4 + 1 = 3$. Two of the three units of input come from the neuron itself.

SECTION 03

The linear autapse

DEF · LINEAR ACTIVATION FUNCTION

The linear (identity) activation function is $g(h) = h$: the rate equals the input, with no threshold and no saturation [LMLF eq. 2.4].

PITFALL · WHY LINEAR IS A SIMPLIFICATION

A real neuron cannot fire at a negative rate, and it cannot fire faster than a few hundred Hz. The linear g ignores both limits. We use it because the resulting equation is linear, so we can solve it and understand it completely. It is a good approximation when the rate changes only a little.

With $g(h) = h$ the autapse equation becomes

$$\tau\dot{r} = -r + Jr + h^{\text{ext}} \qquad \Longrightarrow \qquad \boxed{\tau\dot{r} = -(1-J)\,r + h^{\text{ext}}}$$

TWO FORCES ON r

- $-r$: the leak. Left alone, the rate decays back to zero (Week 1).
- $+Jr$: the feedback. The neuron excites itself in proportion to its own rate.

The whole story of this week is the contest between leak and feedback: who wins depends on whether $J < 1$ or $J > 1$.

This is exactly the equation $\dot{n} = an + I$ from the Week 1 bonus hour, with $a = -(1-J)/\tau$ and constant input $I = h^{\text{ext}}/\tau$. We could solve it right away (we do so in §6). But first we introduce an approach that works *even when the equation cannot be solved*: look for the fixed points and ask whether they are stable.

SECTION 04

Fixed points

DEF · FIXED POINT

A fixed point r^* is a value of r where the dynamics stand still: $\dot{r} = 0$. If the system starts exactly at r^*, it stays there forever [LMLF 2.3.1].

Finding it. Set the right-hand side to zero:

$$0 = -(1-J)\,r^* + h^{\text{ext}} \qquad \Longrightarrow \qquad \boxed{r^* = \frac{h^{\text{ext}}}{1-J}} \qquad (J \neq 1).$$

EXAMPLE · FIXED POINTS FOR $h^{\text{ext}} = 1$

J	$r^* = 1/(1-J)$	COMMENT
0	1	no feedback: the rate just copies the input
0.5	2	feedback doubles the response
0.9	10	strong feedback: ten times the input
0.99	100	close to $J = 1$ the fixed point runs off
1.5	-2	negative! (see below)

FEEDBACK AMPLIFIES

For $0 < J < 1$, $r^* = h^{\text{ext}}/(1-J) > h^{\text{ext}}$: the autapse amplifies its input by the factor $1/(1-J)$. Why: the input raises r, which raises the feedback Jr, which raises r further, and so on. The extra contributions add up to $h^{\text{ext}}(1 + J + J^2 + \dots) = h^{\text{ext}}/(1-J)$, a geometric series that converges only if $J < 1$.

PITFALL · A FIXED POINT IS NOT AUTOMATICALLY WHERE THE SYSTEM GOES

For $J = 1.5$ the formula gives $r^* = -2$. It is a perfectly valid solution of $\dot{r} = 0$, but (as we will see) the system runs *away* from it. Also, a negative rate shows that the linear model has left the range where it describes a real neuron. Finding a fixed point is only half the job; the other half is its stability.

PITFALL · THE BORDER CASE $J = 1$

For $J = 1$ the leak and the feedback cancel exactly: $\tau\dot{r} = h^{\text{ext}}$. If $h^{\text{ext}} \neq 0$ there is *no* fixed point (r grows or shrinks steadily). If $h^{\text{ext}} = 0$ then $\dot{r} = 0$ everywhere: *every* value of r is a fixed point. The book returns to this special case later (§4.2); this week we assume $J \neq 1$.

SECTION 05

Stability

> 5.1 The idea

DEF · STABLE AND UNSTABLE FIXED POINTS

Start the system *near* r^* instead of exactly on it.

- If the dynamics bring r ever closer to r^*, the fixed point is stable.
- If the dynamics push r away from r^*, the fixed point is unstable.



stable: a ball in a valley unstable: a ball on a hilltop

Both balls are at rest (fixed points). A small nudge brings the left ball back but sends the right one rolling away.

WHY STABILITY MATTERS

In the real brain the rate is never exactly at r^*: there is always noise. An unstable fixed point is never seen in practice, because the tiniest disturbance grows. Only stable fixed points describe states the network actually sits in.

> 5.2 Perturbation analysis, step by step

To decide stability, follow a small deviation from the fixed point [LMLF eq. 2.14-2.15].

DEF · PERTURBATION

The perturbation is the distance from the fixed point: $\delta r(t) = r(t) - r^*$. Equivalently $r = r^* + \delta r$.

1. **Substitute** $r = r^* + \delta r$ into $\tau \dot{r} = -r + Jr + h^{\text{ext}}$:

$$\tau \frac{d}{dt}(r^* + \delta r) = -(r^* + \delta r) + J(r^* + \delta r) + h^{\text{ext}}.$$

2. r^* **is a number, not a function of time**, so $\frac{d}{dt} r^* = 0$ and the left side is $\tau\, \dot{\delta r}$.

3. **Regroup the right side** into the part with r^* and the part with δr:

$$\tau\, \dot{\delta r} = \underbrace{\big(-r^* + Jr^* + h^{\text{ext}}\big)}_{=0 \text{ (definition of } r^*)} + \big(-\delta r + J\delta r\big).$$

4. **The first bracket vanishes** because r^* is a fixed point. What remains is

$$\boxed{\tau\, \dot{\delta r} = -(1 - J)\, \delta r}$$

WHAT HAPPENED

The external input h^{ext} has disappeared. It only decides *where* the fixed point is; whether the fixed point is stable depends only on J.

> 5.3 Reading off stability from the sign

Case $J < 1$, so $1 - J > 0$:

- if $\delta r > 0$ (rate above the fixed point) then $\dot{\delta r} < 0$: it comes down;
- if $\delta r < 0$ (rate below) then $\dot{\delta r} > 0$: it goes up.

Either way the perturbation shrinks. The fixed point is stable.

Case $J > 1$, so $1 - J < 0$: now $\dot{\delta r}$ has the *same* sign as δr. Above the fixed point the rate rises further; below it, it falls further. The fixed point is unstable.

STABILITY OF THE LINEAR AUTAPSE

$$J < 1 \;\Rightarrow\; \text{stable} \qquad\qquad J > 1 \;\Rightarrow\; \text{unstable}$$

$J = 1$ is the border between two qualitatively different behaviours.

LEAK VS. FEEDBACK AGAIN

$J < 1$: when the rate goes up by δr, the leak pulls it down by δr but the feedback pushes it up only by $J\delta r < \delta r$. Leak wins, the deviation dies. $J > 1$: the feedback push $J\delta r$ beats the leak, the deviation feeds itself.

> 5.4 How fast? Solving for $\delta r(t)$

$\tau \dot{\delta r} = -(1 - J)\delta r$ is the Week 1 equation $\dot{n} = an$ with $a = -(1 - J)/\tau$, so [LMLF eq. 2.16]

PERTURBATION OVER TIME

$$\delta r(t) = \delta r(0)\, e^{-(1-J)t/\tau}.$$

- $J < 1$: exponential **decay** to 0, i.e. $r(t) \to r^*$.
- $J > 1$: exponential **growth**; r diverges to $+\infty$ or $-\infty$, depending on the sign of $\delta r(0)$.

EXAMPLE · HOW LONG A PERTURBATION LASTS, $\tau = 10$ MS

The perturbation shrinks (or grows) by a factor e every $\tau/|1-J|$:

J	$\tau/\lvert 1-J\rvert$	BEHAVIOUR
0	10 ms	decays (just the leak)
0.5	20 ms	decays, twice as slowly
0.9	100 ms	decays, ten times as slowly
1.5	20 ms	grows by e every 20 ms

The closer J is to 1, the slower everything happens: stronger feedback lets the neuron hold on to a deviation for much longer than its own τ.

SECTION 06

The full solution

Using the Week 1 result for $\dot{n} = an + I$ with constant input (or simply $r = r^* + \delta r$ with $\delta r(t)$ from above):

LINEAR AUTAPSE, SOLVED

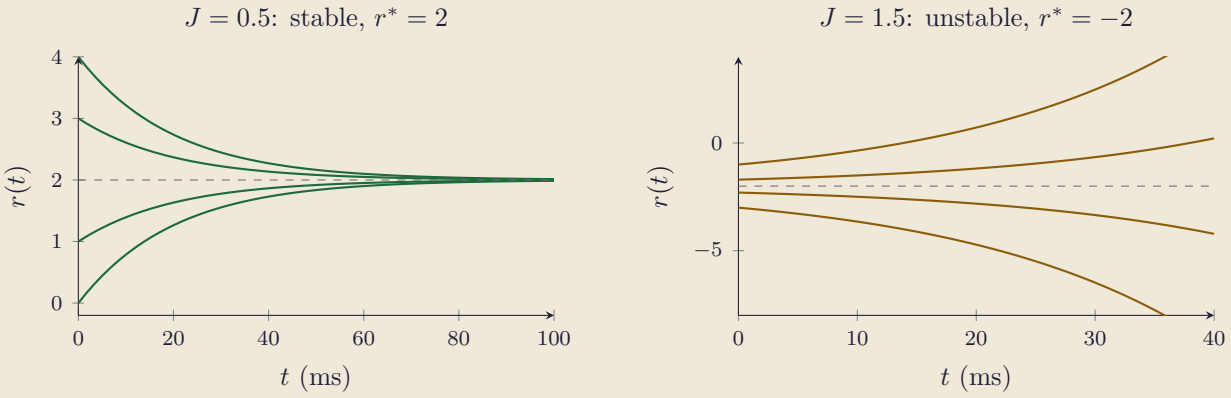
$$r(t) = r^* + \big(r(0) - r^*\big)\, e^{-(1-J)t/\tau}, \qquad r^* = \frac{h^{\text{ext}}}{1-J}.$$

Check. At $t = 0$: $r(0) = r^* + r(0) - r^* = r(0)$ ✓. Differentiate: $\tau\dot{r} = -(1-J)(r(0)-r^*)e^{-(1-J)t/\tau} = -(1-J)(r - r^*) = -(1-J)r + (1-J)r^* = -(1-J)r + h^{\text{ext}}$. ✓

EXAMPLE · STARTING FROM SILENCE, $r(0) = 0$, $h^{\text{ext}} = 1$, $\tau = 10$ MS

t (ms)	0	10	20	40	60	100
$J = 0.5$: $2(1 - e^{-t/20})$	0	0.79	1.26	1.73	1.90	1.99
$J = 0.9$: $10(1 - e^{-t/100})$	0	0.95	1.81	3.30	4.51	6.32
$J = 1.5$: $-2 + 2e^{t/20}$	0	1.30	3.44	12.8	38.2	295

$J = 0.5$ settles at 2 within ∼60 ms. $J = 0.9$ heads for 10 but needs hundreds of ms. $J = 1.5$ starts *above* its fixed point -2, so it is pushed further and further up, without limit.



$r(t)$ from four starting points ($\tau = 10\,\text{ms}$, $h^{\text{ext}} = 1$). Left: every trajectory converges to the dashed fixed point. Right: every trajectory that does not start exactly on the fixed point is driven away from it.

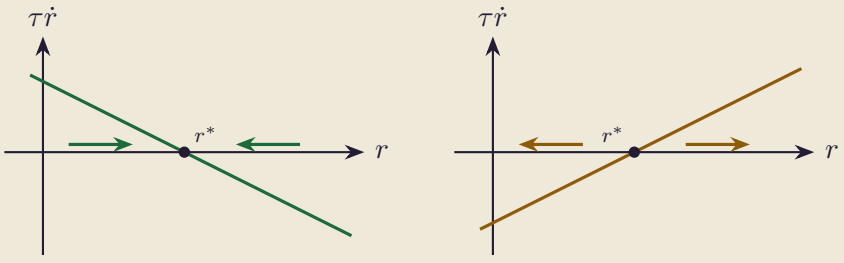
SECTION 07

The geometric view: the vector field

We can see stability *without solving anything*, by plotting $\dot{r}$ as a function of r (the idea from the Week 1 bonus hour).

DEF · VECTOR FIELD ON A LINE

The equation $\tau\dot{r} = F(r)$ assigns a velocity $\dot{r}$ to every value of r: a vector field on the line [LMLF 2.3.1]. Draw $\dot{r}$ against r and put arrows on the r-axis: right where $\dot{r} > 0$, left where $\dot{r} < 0$. The fixed points are where the graph crosses zero.



$J < 1$: slope $-(1-J) < 0$
arrows point *in*: stable

$J > 1$: slope $J - 1 > 0$
arrows point *out*: unstable

$\tau\dot{r} = -(1-J)r + h^{\text{ext}}$ is a straight line in r that crosses zero at r^*. Its slope decides everything.

STABILITY FROM THE SLOPE

A fixed point is stable if the graph of $\dot{r}$ against r crosses zero going **down** (negative slope), and unstable if it crosses going **up** (positive slope). For the linear autapse the slope is $-(1-J)/\tau$: negative exactly when $J < 1$.

WHY THIS IS POWERFUL

This picture needs only the *sign* of $\dot{r}$ around r^*. It works for equations we cannot solve, which will be the usual case once g is not linear.

SECTION 08

Connection to Week 1: it is the eigenvalue

In the TA session of Week 1 we saw that $\dot{\vec{x}} = M\vec{x}$ converges iff all eigenvalues of M are negative. The perturbation equation

$$\dot{\delta r} = \underbrace{-\frac{1-J}{\tau}}_{\text{a } 1\times 1 \text{ “matrix”}} \delta r$$

is the one-dimensional case: its only eigenvalue is $-(1-J)/\tau$, negative exactly when $J < 1$. Stability of a fixed point = the sign of the eigenvalue of the perturbation dynamics. For networks of many neurons the same recipe gives a matrix, and Week 1's eigenvalue tools take over.

SECTION 09

A limitation of the linear model

PITFALL · DIVERGENCE IS NOT BIOLOGY

For $J > 1$ the linear autapse fires at a rate that grows without bound (or goes to $-\infty$). Real neurons *saturate*: their rate cannot exceed a few hundred Hz, and it cannot go below zero. This is a limitation of the linear g, not of the autapse itself. It is the reason the book next replaces $g(h) = h$ by a saturating activation function [LMLF 2.3.1].

SECTION 10

Practice problems with full solutions

EXAMPLE · P1: FIXED POINT AND STABILITY

$\tau\dot{r} = -r + 0.8r + 3$. Find r^*, decide stability, and give $r(t)$ for $r(0) = 0$, $\tau = 10\,\text{ms}$.

Solution. $J = 0.8$, $h^{\text{ext}} = 3$: $r^* = 3/0.2 = 15$. $J < 1$, so stable. $r(t) = 15(1 - e^{-0.2t/10}) = 15(1 - e^{-t/50})$: the time scale is 50 ms, five times τ.

EXAMPLE · P2: WHICH J GIVES A TARGET RATE?

With $h^{\text{ext}} = 2$, what J makes the stable fixed point $r^* = 8$?

Solution. $8 = 2/(1 - J) \Rightarrow 1 - J = 0.25 \Rightarrow J = 0.75$. Since $0.75 < 1$ it is stable. ✓

EXAMPLE · P3: PERTURBATION BY HAND

$J = 1.2$, $h^{\text{ext}} = -0.4$, $\tau = 10\,\text{ms}$. Find r^*. The rate starts at $r(0) = 2.1$; what is it after 50 ms?

Solution. $r^* = -0.4/(1 - 1.2) = -0.4/(-0.2) = 2$. $J > 1$: unstable. $\delta r(0) = 0.1$, and $\delta r(t) = 0.1\, e^{0.2t/10} = 0.1\, e^{t/50}$. At $t = 50$: $\delta r = 0.1e \approx 0.27$, so $r \approx 2.27$. The rate drifts away from 2, slowly at first and then faster and faster.

EXAMPLE · P4: VECTOR FIELD

Sketch $\tau\dot{r}$ against r for $\tau\dot{r} = -0.5r + 1$ and use it to say what happens from $r(0) = 5$.

Solution. A line with slope -0.5 crossing zero at $r^* = 2$. At $r = 5$: $\tau\dot{r} = -1.5 < 0$, so r decreases, and it keeps decreasing until it reaches 2. The arrows point towards 2 from both sides: stable.

EXAMPLE · P5: THE BORDER

What happens for $J = 1$, $h^{\text{ext}} = 0.5$, $\tau = 10\,\text{ms}$, $r(0) = 0$?

Solution. $\tau\dot{r} = 0.5$, so $\dot{r} = 0.05$ per ms: $r(t) = 0.05t$, a straight line. There is no fixed point; the rate grows steadily, not exponentially.

SECTION 11

Cheat sheet

WEEK 02 ON ONE PAGE

- **Autapse**: a neuron's synapse onto itself, strength J. $\tau\dot{r} = -r + g(Jr + h^{\text{ext}})$. Excitatory: $J > 0$ (positive feedback).
- **Linear** $g(h) = h$: $\tau\dot{r} = -(1 - J)r + h^{\text{ext}}$. Leak $-r$ vs. feedback $+Jr$.
- **Fixed point**: $\dot{r} = 0$. Linear autapse: $r^* = h^{\text{ext}}/(1 - J)$ $(J \neq 1)$. For $0 < J < 1$, an amplification by $1/(1 - J)$.
- **Stable**: nearby states return; **unstable**: nearby states run away. Only stable fixed points are seen in practice.
- **Perturbation**: $\delta r = r - r^*$; $\tau\dot{\delta r} = -(1 - J)\delta r$; $\delta r(t) = \delta r(0)e^{-(1-J)t/\tau}$. h^{ext} drops out.
- $J < 1$: stable, decay time $\tau/(1 - J)$ (slower as $J \to 1$). $J > 1$: unstable, divergence. $J = 1$: border (no fixed point, or a whole line of them).
- **Full solution**: $r(t) = r^* + (r(0) - r^*)e^{-(1-J)t/\tau}$.
- **Vector field**: plot $\dot{r}$ vs. r; crossing zero going down = stable, going up = unstable.

- Stability = sign of the eigenvalue of the perturbation dynamics (here $-(1-J)/\tau$), as in Week 1.